10.3 Causal Models for Density Estimation

In this section, we present a number of models in which representational learning is achieved through density estimation. The mixture of Gaussians and factor analysis models that we have already mentioned are examples of invertible generative models with probabilistic recognition. K-means is a limiting case of mixture of Gaussians with deterministic recognition, and principal components analysis is a limiting case of factor analysis with deterministic recognition. We consider two other models with deterministic recognition: independent components analysis, which is invertible; and sparse coding, which is noninvertible. Our final example, the Helmholtz machine, is noninvertible with probabilistic recognition. The Boltzmann machine, discussed in chapters 7 and 8, is an additional example that is closely related to the causal models discussed here. We summarize and interpret general properties of representations derived from causal models at the end of the chapter. The table in the appendix summarizes the generative and recognition distributions and the learning rules for all the models we discuss.

Mixture of Gaussians

The model applied in the introduction to the data in figure 10.1A is a mixture of Gaussians model. That example involved two causes and two Gaussian distributions, but we now generalize this to N_v causes, each associated with a separate Gaussian distribution. The model is defined by the probability distributions

$$P[v;\mathcal{G}] = \gamma_v \quad \text{and} \quad p[\mathbf{u}|v;\mathcal{G}] = \mathcal{N}(\mathbf{u};\mathbf{g}_v,\Sigma_v)\,, \tag{10.17}$$

where v takes N_v values representing the different causes and, for an N_u component input vector,

$$\mathcal{N}(\mathbf{u};\mathbf{g},\Sigma) = \frac{1}{(2\pi\Sigma)^{N_u/2}}\exp\left(-\frac{|\mathbf{u}-\mathbf{g}|^2}{2\Sigma}\right) \tag{10.18}$$

is a Gaussian distribution with mean $\mathbf{g}$ and variances for the individual components equal to Σ. The function $\mathcal{F}(Q,\mathcal{G})$ for this model is given by an expression similar to equation 10.14 (with slightly different factors if $N_u \neq 2$), leading to the M-phase learning rules given in the appendix. Once the generative model has been optimized, the recognition distribution is constructed from equation 10.3 as

$$P[v|\mathbf{u};\mathcal{G}] = \frac{\gamma_v\mathcal{N}(\mathbf{u};\mathbf{g}_v,\Sigma_v)}{\sum_{v'}\gamma_{v'}\mathcal{N}(\mathbf{u};\mathbf{g}_{v'},\Sigma_{v'})}\,. \tag{10.19}$$

K-Means Algorithm

A special case of mixture of Gaussians can be derived in the limit that the variances of the Gaussians are equal and tend toward 0, $\Sigma_v = \Sigma \to 0$. We

discuss this limit for two clusters, as in figure 10.1. When Σ is extremely small, the recognition distribution $P[v|\mathbf{u};\mathcal{G}]$ of equation 10.19 degenerates because it takes essentially two values, 0 or 1, depending on whether $\mathbf{u}$ is closer to one cluster or the other. This provides a deterministic, rather than a probabilistic, classification of $\mathbf{u}$. In the degenerate case, EM consists of choosing two random values for the centers of the two cluster distributions, and then repeatedly finding all the inputs $\mathbf{u}$ that are closest to a given center $\mathbf{g}_v$, and then moving $\mathbf{g}_v$ to the average of these points. This is called the K-means algorithm (with $K = 2$ for two clusters). The mixing proportions γ_v do not play an important role for the K-means algorithm. New input points are recognized as belonging to the clusters to which they are closest.

Factor Analysis

The model used in figure 10.3 is an example of factor analysis. In general, factor analysis involves a continuous vector of causes, $\mathbf{v}$, drawn from a Gaussian prior distribution, and uses a Gaussian generative distribution with a mean that depends linearly on $\mathbf{v}$. We assume that the distribution $p[\mathbf{u}]$ has a mean of 0 (nonzero means can be accommodated simply by shifting the input data). The defining distributions for factor analysis are thus

$$p[\mathbf{v};\mathcal{G}] = \mathcal{N}(\mathbf{v};\mathbf{0},1) \quad \text{and} \quad p[\mathbf{u}|\mathbf{v};\mathcal{G}] = \mathcal{N}(\mathbf{u};\mathbf{G}\cdot\mathbf{v},\mathbf{\Sigma}), \tag{10.20}$$

where the extension of equation 10.18, expressed in terms of the mean $\mathbf{g}$ and covariance matrix $\mathbf{\Sigma}$, is

$$\mathcal{N}(\mathbf{u};\mathbf{g},\mathbf{\Sigma}) = \frac{1}{((2\pi)^{N_u}|\det\mathbf{\Sigma}|)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{u}-\mathbf{g})\cdot\mathbf{\Sigma}^{-1}\cdot(\mathbf{u}-\mathbf{g})\right). \tag{10.21}$$

The expression $|\det\mathbf{\Sigma}|$ indicates the (absolute) value of the determinant of $\mathbf{\Sigma}$. In factor analysis, $\mathbf{\Sigma}$ is taken to be diagonal, $\mathbf{\Sigma} = \mathrm{diag}(\Sigma_1,\ldots,\Sigma_{N_u})$ (see the Mathematical Appendix), with all the diagonal elements strictly positive, so its inverse is simply $\mathbf{\Sigma}^{-1} = \mathrm{diag}(1/\Sigma_1,\ldots,1/\Sigma_{N_u})$ and $|\det\mathbf{\Sigma}| = \Sigma_1\Sigma_2\ldots\Sigma_{N_u}$.

Because $\mathbf{\Sigma}$ is diagonal, the individual components of $\mathbf{v}$ are mutually independent. Thus, any correlations between the components of $\mathbf{u}$ must arise from the mean values $\mathbf{G}\cdot\mathbf{v}$ of the generative distribution. To be well specified, the model requires $\mathbf{v}$ to have fewer dimensions than $\mathbf{u}$ ($N_v < N_u$). In terms of heuristics, factor analysis seeks a relatively small number of independent causes that account, in a linear manner, for collective Gaussian structure in the inputs.

The recognition distribution for factor analysis has the Gaussian form

$$p[\mathbf{v}|\mathbf{u};\mathcal{G}] = \mathcal{N}(\mathbf{v};\mathbf{W}\cdot\mathbf{u},\mathbf{\Psi}), \tag{10.22}$$

where expressions for $\mathbf{W}$ and $\mathbf{\Psi}$ are given in the appendix. These do not depend on the input $\mathbf{u}$, so factor analysis involves a linear relation between the input and the mean of the recognition distribution. EM, as applied to an invertible model, can be used to adjust $\mathcal{G} = (\mathbf{G}, \mathbf{\Sigma})$ on the basis of the input data. The resulting learning rules are given in the appendix. For the case of a single cause v, these reduce to equation 10.5.

In this case, we can understand the goal of density estimation in an additional way. By direct calculation, as in equation 10.1, the marginal distribution for $\mathbf{u}$ is

$$p[\mathbf{u}; \mathcal{G}] = \mathcal{N}(\mathbf{u}; \mathbf{0}, \mathbf{G} \cdot \mathbf{G}^{\mathrm{T}} + \mathbf{\Sigma}), \tag{10.23}$$

where $[\mathbf{G}^{\mathrm{T}}]_{ab} = [\mathbf{G}]_{ba}$ and $[\mathbf{G} \cdot \mathbf{G}^{\mathrm{T}}]_{ab} = \sum_c G_{ac} G_{bc}$ (see the Mathematical Appendix). Maximum likelihood density estimation requires determining the $\mathcal{G}$ that makes $\mathbf{G} \cdot \mathbf{G}^{\mathrm{T}} + \mathbf{\Sigma}$ match, as closely as possible, the covariance matrix of the input distribution.

Principal Components Analysis

In the same way that setting the parameters Σ_v to 0 in the mixture of Gaussians model leads to the K-means algorithm, setting all the variances in factor analysis to 0 leads to another well-known method, principal components analysis (which we also discuss in chapter 8). To see this, consider the case of a single factor. This means that v is a single number, and that the mean of the distribution $p[\mathbf{u}|v; \mathcal{G}]$ is $v\mathbf{g}$, where the vector $\mathbf{g}$ replaces the matrix $\mathbf{G}$ of the general case. The elements of the diagonal matrix $\mathbf{\Sigma}$ are set to a single variance Σ, which we shrink to 0.

As $\Sigma \to 0$, the Gaussian distribution $p[\mathbf{u}|v; \mathcal{G}]$ in equation 10.20 approaches a δ function (see the Mathematical Appendix), and it can generate only the single vector $\mathbf{u}(v) = v\mathbf{g}$ from cause v. Similarly, the recognition distribution of equation 10.22 becomes a δ function, making the recognition process deterministic with $v(\mathbf{u}) = \mathbf{W} \cdot \mathbf{u}$ given by the mean of the recognition distribution of equation 10.22. Using the expression for $\mathbf{W}$ in the appendix in the limit $\Sigma \to 0$, we find

$$v(\mathbf{u}) = \frac{\mathbf{g} \cdot \mathbf{u}}{|\mathbf{g}|^2}. \tag{10.24}$$

This is the result of the E phase of EM. In the M phase, we maximize

$$\mathcal{F}(v(\mathbf{u}), \mathcal{G}) = \langle \ln p[v(\mathbf{u}), \mathbf{u}; \mathcal{G}] \rangle = K - \frac{N_u \ln \Sigma}{2} - \left\langle \frac{v^2(\mathbf{u})}{2} + \frac{|\mathbf{u} - \mathbf{g}v(\mathbf{u})|^2}{2\Sigma} \right\rangle \tag{10.25}$$

with respect to $\mathbf{g}$, without changing the expression for $v(\mathbf{u})$. Here, K is a term independent of $\mathbf{g}$ and Σ. In this expression, the only term that depends on $\mathbf{g}$ is proportional to $|\mathbf{u} - \mathbf{g}v(\mathbf{u})|^2$. Minimizing this in the M

phase produces a new value of $\mathbf{g}$ given by

$$\mathbf{g} = \frac{\langle v(\mathbf{u})\mathbf{u}\rangle}{\langle v^2(\mathbf{u})\rangle} . \tag{10.26}$$

This depends only on the covariance matrix of the input distribution, as does the more general form given in the appendix. Under EM, equations 10.24 and 10.26 are alternated until convergence.

For principal components analysis, we can say more about the value of $\mathbf{g}$ at convergence. We consider the case $|\mathbf{g}|^2 = 1$ because we can always multiply $\mathbf{g}$ and divide $v(\mathbf{u})$ by the same factor to make this true without affecting the dominant term in $\mathcal{F}(v(\mathbf{u}), \mathcal{G})$ as $\Sigma \to 0$. Then, the $\mathbf{g}$ that maximizes this dominant term must minimize

$$\left\langle |\mathbf{u} - \mathbf{g}(\mathbf{g}\cdot\mathbf{u})|^2 \right\rangle = \left\langle |\mathbf{u}|^2 - (\mathbf{g}\cdot\mathbf{u})^2 \right\rangle . \tag{10.27}$$

Here, we have used expression 10.24 for $v(\mathbf{u})$. Minimizing 10.27 with respect to $\mathbf{g}$, subject to the constraint $|\mathbf{g}|^2 = 1$, gives the result that $\mathbf{g}$ is the eigenvector of the covariance matrix $\langle \mathbf{uu} \rangle$ with maximum eigenvalue. This is just the principal component vector and is equivalent to finding the vector of unit length with the largest possible average projection onto $\mathbf{u}$. Note that there are ways other than EM for finding eigenvectors of this matrix.

The argument we have given shows that principal components analysis is a degenerate form of factor analysis. This is also true if more than one factor is considered, although maximizing $\mathcal{F}$ constrains the projections $\mathbf{G}\cdot\mathbf{u}$ and therefore is sufficient only to force $\mathbf{G}$ to represent the principal components subspace of the data. The same subspace emerges from full factor analysis provided that the variances of all the factors are equal, even when they are nonzero.

Figure 10.4 illustrates an important difference between factor analysis and principal components analysis. In this example, $\mathbf{u}$ is a three-component input vector, $\mathbf{u} = (u_1, u_2, u_3)$. Just as in figure 10.3, samples of input data were generated on the basis of a "true" cause, v_{true} according to

$$u_b = v_{\text{true}} + \epsilon_b , \tag{10.28}$$

where ϵ_b represents the noise added to component b of the input. Input data points were generated from this equation by choosing a value of v_{true} from a Gaussian distribution with mean 0 and variance 1, and values of ϵ_b from independent Gaussian distributions with 0 means. The variances of the distributions for ϵ_1, ϵ_2, and ϵ_3 were all are equal to 0.25 in figures 10.4A and 10.4B. However, in figures 10.4C and 10.4D, the variance for ϵ_3 is much larger (equal to 9), as if the third sensor was much noisier than the other two sensors. The graphs in figure 10.4 show the mean of the values of v extracted from sample inputs by factor analysis, or the value of v for principal components analysis, as a function of the true value used to generate the data. Perfect extraction of the underlying cause would have $v = v_{\text{true}}$, but this is impossible in this case because of the noise. The

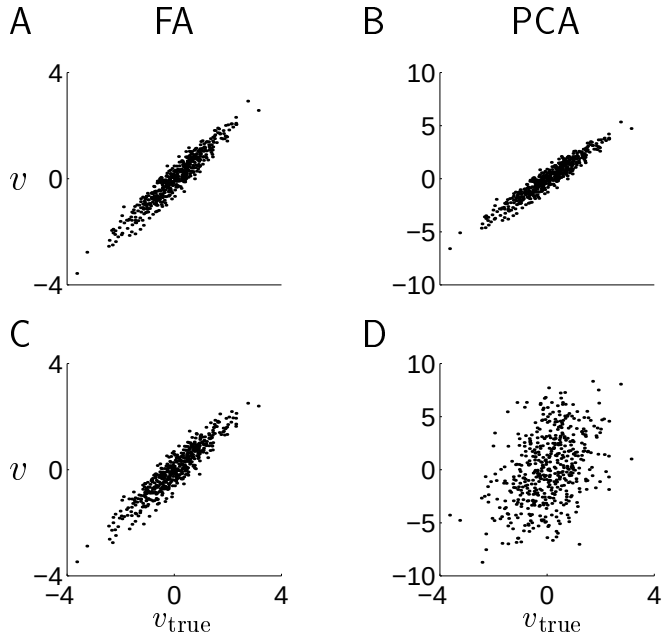

Figure 10.4 Factor analysis (FA) and principal components analysis (PCA) applied to 500 samples of noisy input reflecting a single underlying cause v_{true}. For A and B, $\langle u_i u_j \rangle = 1 + 0.25\delta_{ij}$, whereas for C and D, one sensor is corrupted by independent noise with standard deviation 3 rather than 0.5. The plots compare the values of the true cause v_{true} and the cause v inferred by the model.

best we can expect is for the v values to be well correlated with the values of v_{true}. When the input components are equally variable (figure 10.4A and 10.4B), this is indeed what happens for both factor and principal components analysis. However, when u_3 is much more variable than the other components, principal components analysis (figure 10.4D) is fooled by the extra variance and finds a cause v that does not correlate very well with v_{true}. On the other hand, factor analysis (figure 10.4C) is affected only by the covariance between the input components and not by their individual variances (which are absorbed into $\mathbf{\Sigma}$), so the cause it finds is not significantly perturbed (merely slightly degraded) by the added sensor noise.

In chapter 8, we noted that principal components analysis maximizes the mutual information between the input and output under the assumption of a linear Gaussian model. This property, and the fact that principal components analysis minimizes the reconstruction error of equation 10.27, have themselves been suggested as goals for representational learning. We have now shown how they are also related to density estimation.

Both principal components analysis and factor analysis produce a marginal distribution $p[\mathbf{u}; \mathcal{G}]$ that is Gaussian. If the actual input distribution $p[\mathbf{u}]$ is non-Gaussian, the best that these models can do is to match the mean and covariance of $p[\mathbf{u}]$; they will fail to match higher-order moments. If the input is whitened to increase coding efficiency, as discussed

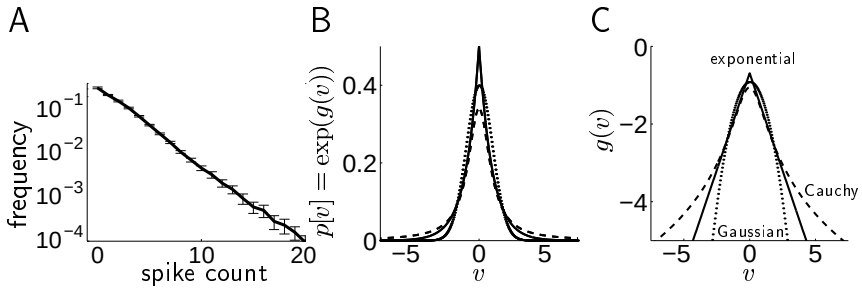

Figure 10.5 Sparse distributions. (A) Log frequency distribution for the activity of a macaque IT cell in response to video images. The fraction of times that various numbers of spikes appeared in a spike-counting window is plotted against the number of spikes. The size of the window was adjusted so that, on average, there were two spikes per window. (B) Three distributions $p[v] = \exp(g(v))$: double exponential ($g(v) = -|v|$, solid, kurtosis = 3); Cauchy ($g(v) = -\ln(1 + v^2)$, dashed, kurtosis = ∞); and Gaussian ($g(v) = -v^2/2$, dotted, kurtosis = 0). (C) The logarithms of the same three distributions. (A adapted from Baddeley et al., 1997.)

in chapter 4, so that the covariance matrix $\langle \mathbf{uu} \rangle$ is equal to the identity matrix, neither method will extract any structure at all from the input data. By contrast, the generative models discussed in the following sections produce non-Gaussian marginal distributions and attempt to account for structure in the input data beyond merely covariance (and the mean).

Sparse Coding

The v values in response to input in factor and principal components analysis tend to be Gaussian distributed. If we attempt to relate such causal variables to the activities of cortical neurons, we find a discrepancy, because the activity distributions of cortical cells in response to natural inputs are not Gaussian. Figure 10.5A shows an example of the distribution of the numbers of spikes counted within a particular time window for a neuron in the inferotemporal (IT) area of the macaque brain recorded while a monkey freely viewed television shows. The distribution is close to being exponential. This means that the neurons are most likely to fire a small number of spikes in the counting interval, but that they can occasionally fire a large number of spikes.

sparse distributions

Distributions that generate values for the components of $\mathbf{v}$ close to 0 most of the time, but occasionally far from 0, are called sparse. Intuitively, sparse distributions are more likely than Gaussians of the same mean and variance to generate values near 0, and also more likely to generate values far from 0. These occasional high values can convey substantial information. Distributions with this character are also called heavy-tailed. Figures 10.5B and 10.5C compare two sparse distributions with a Gaussian distribution.

More formally, sparseness has been defined in a variety of ways. Sparseness of a distribution is sometimes linked to a high value of a measure called kurtosis. Kurtosis of a distribution $p[v]$ is defined as *kurtosis*

$$k = \frac{\int dv\, p[v](v-\overline{v})^4}{\left(\int dv\, p[v](v-\overline{v})^2\right)^2} - 3 \quad \text{with} \quad \overline{v} = \int dv\, p[v]v\,, \tag{10.29}$$

and it takes the value 0 for a Gaussian distribution. Positive values of k are taken to imply sparse distributions, which are also called super-Gaussian or leptokurtotic. Distributions with $k < 0$ are called sub-Gaussian or platykurtotic. This is a slightly different definition of sparseness from being heavy-tailed.

A sparse representation over a large population of neurons might more naturally be defined as one in which each input is encoded by a small number of the neurons in the population. Unfortunately, identifying this form of sparseness experimentally is difficult.

Sparse coding can arise in generative models that have sparse prior distributions over causes. Unlike factor analysis and principal components analysis, sparse coding does not stress minimizing the number of representing units (i.e., components of $\mathbf{v}$), and sparse representations may require large numbers of units. This is not a disadvantage for modeling the visual system because representations in visual areas are indeed greatly expanded at various steps along the pathway. For example, there are around 40 cells in primary visual cortex for each cell in the visual thalamus. Downstream processing can benefit greatly from sparse representations because, for one thing, they minimize interference between different patterns of input.

Because they employ Gaussian priors, factor analysis and principal components analysis do not generate sparse representations. The mixture of Gaussians model is extremely sparse because each input is represented by a single cause. This may be reasonable for relatively simple input patterns, but for complex stimuli such as images, we seek something between these extremes. Olshausen and Field (1996, 1997) suggested such a model by considering a nonlinear version of factor analysis. In this model, the distribution of $\mathbf{u}$ given $\mathbf{v}$ is Gaussian with a diagonal covariance matrix, as for factor analysis, but the prior distribution over causes is sparse. Defined in terms of a function $g(v)$ (as in figure 10.5), the model has

$$p[\mathbf{v};\mathcal{G}] \propto \prod_{a=1}^{N_v} \exp(g(v_a)) \quad \text{and} \quad p[\mathbf{u}|\mathbf{v};\mathcal{G}] = \mathcal{N}(\mathbf{u};\mathbf{G}\cdot\mathbf{v},\mathbf{\Sigma})\,. \tag{10.30}$$

The prior $p[\mathbf{v};\mathcal{G}]$ should be normalized so that its integral over $\mathbf{v}$ is 1, but we omit the normalization factor to simplify the equations.

The prior $p[\mathbf{v};\mathcal{G}]$ in equation 10.30 makes the components of $\mathbf{v}$ mutually independent because it is a product. If we took $g(v) = -v^2$, $p[\mathbf{v};\mathcal{G}]$ would be Gaussian (dotted lines in figures 10.5B and 10.5C), and the model would

perform factor analysis. An example of a function that provides a sparse prior is $g(v) = -\alpha|v|$. This generates a double exponential distribution (solid lines in figures 10.5B and 10.5C) similar to the activity distribution in figure 10.5A. Another commonly used form is

double exponential distribution

$$g(v) = -\ln(\beta^2 + v^2) \tag{10.31}$$

Cauchy distribution

with β a constant, which generates a Cauchy distribution (dashed lines in figures 10.5B and 10.5C).

For $g(v)$ such as equation 10.31, it is difficult to compute the recognition distribution $p[\mathbf{v}|\mathbf{u}; \mathcal{G}]$ exactly. This makes the sparse model noninvertible. Olshausen and Field chose a deterministic approximate recognition model. Thus, EM consists of finding $\mathbf{v}(\mathbf{u})$ during the E phase, and using it to adjust the parameters $\mathcal{G}$ during the M phase. To simplify the discussion, we make the covariance matrix of the generative model proportional to the identity matrix, $\mathbf{\Sigma} = \Sigma\mathbf{I}$. The function to be maximized is then

$$\mathcal{F}(\mathbf{v}(\mathbf{u}), \mathcal{G}) = \left\langle -\frac{1}{2\Sigma}|\mathbf{u} - \mathbf{G}\cdot\mathbf{v}(\mathbf{u})|^2 + \sum_{a=1}^{N_v} g(v_a(\mathbf{u}))\right\rangle + K, \tag{10.32}$$

where K is a term that is independent of $\mathbf{G}$ and $\mathbf{v}$. For convenience in discussing the EM procedure, we further take $\Sigma = 1$ and do not allow it to vary. Similarly, we assume that β in equation 10.31 is predetermined and held fixed. Then, $\mathcal{G}$ consists only of the matrix $\mathbf{G}$.

The E phase of EM involves maximizing $\mathcal{F}$ with respect to $\mathbf{v}(\mathbf{u})$ for every $\mathbf{u}$. This leads to the conditions (for all a)

$$\sum_{b=1}^{N_u}[\mathbf{u} - \mathbf{G}\cdot\mathbf{v}(\mathbf{u})]_b G_{ba} + g'(v_a) = 0\,. \tag{10.33}$$

The prime on $g(v_a)$ indicates a derivative. One way to solve this equation is to let $\mathbf{v}$ evolve over time according to the equation

$$\tau_{\mathbf{v}}\frac{dv_a}{dt} = \sum_{b=1}^{N_u}[\mathbf{u} - \mathbf{G}\cdot\mathbf{v}(\mathbf{u})]_b G_{ba} + g'(v_a)\,, \tag{10.34}$$

where $\tau_{\mathbf{v}}$ is an appropriate time constant. This equation changes $\mathbf{v}$ so that it asymptotically approaches a value $\mathbf{v} = \mathbf{v}(\mathbf{u})$ that satisfies equation 10.33 and sets the right side of equation 10.34 to 0. We assume that the evolution of $\mathbf{v}$ according to equation 10.34 is carried out long enough during the E phase for this to happen. This process is guaranteed to find only a local, not a global, maximum of $\mathcal{F}$, and it is not guaranteed to find the same local maximum on each iteration.

Equation 10.34 resembles the equation used in chapter 7 for a firing-rate network model. The term $\sum_b u_b G_{ba}$, which can be written in vector form as $\mathbf{G}^{\mathrm{T}}\cdot\mathbf{u}$, acts as the total input arising from units with activities $\mathbf{u}$ fed through a feedforward coupling matrix $\mathbf{G}^{\mathrm{T}}$. The term $-\sum_b[\mathbf{G}\cdot\mathbf{v}]_b G_{ba}$ can

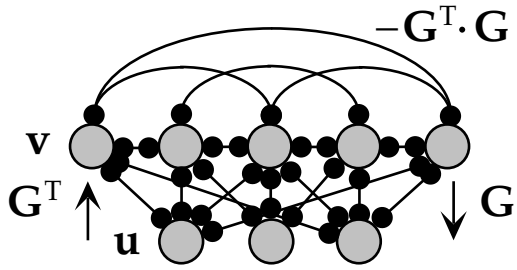

Figure 10.6 A network for sparse coding. This network reproduces equation (10.34), using recurrent weights $-\mathbf{G}^{\mathrm{T}}\cdot\mathbf{G}$ in the $\mathbf{v}$ layer and weights connecting the input units to this layer that are given by the transpose of the matrix $\mathbf{G}$. The reverse connections from the $\mathbf{v}$ layer to the input layer indicate how the mean of the recognition distribution is computed.

be interpreted as a recurrent coupling of the $\mathbf{v}$ units through the matrix $-\mathbf{G}^{\mathrm{T}}\cdot\mathbf{G}$. Finally, the term $g'(v_a)$ plays the same role as the term $-v_a$ that would appear in the rate equations of chapter 7. If $g'(v) \neq -v$, this can be interpreted as a modified form of firing-rate dynamics. Figure 10.6 shows the resulting network. The feedback connections from the $\mathbf{v}$ units to the input units that determine the mean of the generative distribution, $\mathbf{G}\cdot\mathbf{v}$ (equation 10.30), are also shown in this figure.

After $\mathbf{v}(\mathbf{u})$ has been determined during the E phase of EM, a delta rule (chapter 8) is used during the M phase to modify $\mathbf{G}$ and improve the generative model. The full learning rule is given in the appendix. The delta rule follows from maximizing $\mathcal{F}(\mathbf{v}(\mathbf{u}), \mathcal{G})$ with respect to $\mathbf{G}$. A complication arises here because the matrix $\mathbf{G}$ always appears multiplied by $\mathbf{v}$. This means that the bias toward small values of v_a imposed by the prior can be effectively neutralized by scaling up $\mathbf{G}$. This complication results from the approximation of deterministic recognition. To prevent the weights from growing without bound, constraints are applied on the lengths of the generative weights for each cause, $\sum_b G_{ba}^2$, to encourage the variances of all the different v_a to be approximately equal (see the appendix). Further, it is conventional to precondition the inputs before learning by whitening them so that $\langle\mathbf{u}\rangle = 0$ and $\langle\mathbf{u}\mathbf{u}\rangle = \mathbf{I}$. This typically makes learning faster, and it also ensures that the network is forced to find statistical structure beyond second order that would escape simpler methods such as factor analysis or principal components analysis. In the case that the input is created by sampling (e.g., pixelating an image), more sophisticated forms of preconditioning can be used to remove the resulting artifacts.

Applying the sparse coding model to inputs coming from the pixel intensities of small square patches of monochrome photographs of natural scenes leads to selectivities that resemble those of cortical simple cells. Before studying this result, we need to specify how the selectivities of generative models, such as the sparse coding model, are defined. The selectivities of sensory neurons are typically described by receptive fields, as in chapter 2. For a causal model, one definition of a receptive field for unit a is the set of inputs $\mathbf{u}$ for which v_a is likely to take large values. However, it may be

projective field

impossible to construct receptive fields by averaging over these inputs in nonlinear models, such as sparse coding models. Furthermore, generative models are most naturally characterized by projective fields rather than receptive fields. The projective field associated with a particular cause v_a can be defined as the set of inputs that it frequently generates. This consists of all the $\mathbf{u}$ values for which $P[\mathbf{u}|v_a; \mathcal{G}]$ is sufficiently large when v_a is large. For the model of figure 10.1, the projective fields are simply the circles in figure 10.1C. It is important to remember that projective fields can be quite different from receptive fields.

Projective fields for the Olshausen and Field model trained on natural scenes are shown in figure 10.7A, with one picture for each component of $\mathbf{v}$. In this case, the projective field for v_a is simply the matrix elements G_{ab} plotted for all b values. In figure 10.7A, the index b is plotted over a two-dimensional grid representing the location of the input u_b within the visual field. The projective fields form a Gabor-like representation for images, covering a variety of spatial scales and orientations. The resemblance of this representation to the receptive fields of simple cells in primary visual cortex is quite striking, although these are the projective fields of the model, not its receptive fields. Unfortunately, there is no simple form for the receptive fields of the $\mathbf{v}$ units. Figure 10.7B compares the projective field of one unit with receptive fields determined by presenting either dots or gratings as inputs and recording the responses. The responses to the dots directly determine the receptive field, while responses to the gratings directly determine the Fourier transform of the receptive field. Differences between the receptive fields calculated on the basis of these two types of input are evident in the figure. In particular, the receptive field computed from gratings shows more spatial structure than the one mapped by dots. Nevertheless, both show a resemblance to the projective field and to a typical simple-cell receptive field.

In a generative model, projective fields are associated with the causes underlying the visual images presented during training. The fact that the causes extracted by the sparse coding model resemble Gabor patches within the visual field is somewhat strange from this perspective. It is difficult to conceive of images as arising from such low-level causes, instead of causes couched in terms of objects within the images, for example. From the perspective of good representation, causes that are more like objects and less like Gabor patches would be more useful. To put this another way, although the prior distribution over causes biased them toward mutual independence, the causes produced by the recognition model in response to natural images are not actually independent. This is due to the structure in images arising from more complex objects than bars and gratings. It is unlikely that this high-order structure can be extracted by a model with only one set of causes. It is more natural to think of causes in a hierarchical manner, with causes at a higher level accounting for structure in the causes at a lower level. The multiple representations in areas along the visual pathway suggests such a hierarchical scheme, but the corresponding models are still in the rudimentary stages of development.

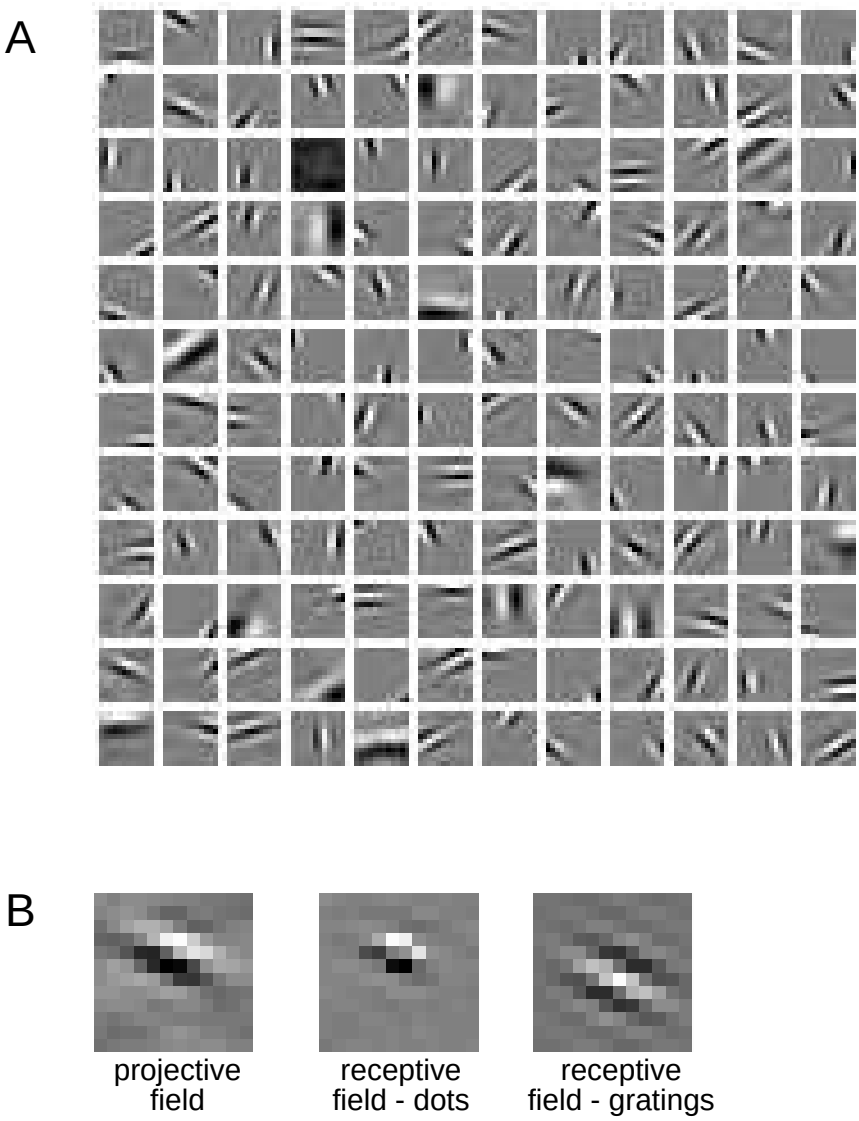

Figure 10.7 Projective and receptive fields for a sparse coding network with $N_u = N_v = 144$. (A) Projective fields $\mathbf{G}_{ab}$ with a indexing representational units (the components of $\mathbf{v}$), and b indexing input units $\mathbf{u}$ on a 12×12 pixel grid. Each box represents a different a value, and the b values are represented within the box by the corresponding input location. Weights are represented by the gray-scale level, with gray indicating 0. (B) The relationship between projective and receptive fields. The left panel shows the projective field of one of the units in A. The middle and right panels show its receptive field mapped using inputs generated by dots and by gratings, respectively. (Adapted from Olshausen & Field, 1997.)

Independent Components Analysis

As for the case of the mixtures of Gaussians model and factor analysis, an interesting model emerges from sparse coding as $\Sigma \to 0$. In this limit, the generative distribution (equation 10.30) approaches a δ function and always generates $\mathbf{u}(\mathbf{v}) = \mathbf{G} \cdot \mathbf{v}$. Under the additional restriction that there are as many causes as inputs, the approximation we used for the sparse coding model of making the recognition distribution deterministic becomes exact, and the recognition distribution that maximizes $\mathcal{F}$ is

$$\mathcal{Q}\left[\mathbf{v}; \mathbf{u}\right] = |\det \mathbf{W}|^{-1} \delta(\mathbf{u} - \mathbf{W}^{-1} \cdot \mathbf{v}) \,, \tag{10.35}$$

where $\mathbf{W} = \mathbf{G}^{-1}$ is the matrix inverse of the generative weight matrix. The factor $|\det \mathbf{W}|$ comes from the normalization condition on Q, $\int d\mathbf{v}\, Q(\mathbf{v}; \mathbf{u}) = 1$. At the maximum with respect to Q, the function $\mathcal{F}$ is

$$\mathcal{F}\left(\mathcal{Q}\,, \mathcal{G}\right) = \left\langle -\frac{1}{2\Sigma} |\mathbf{u} - \mathbf{G} \cdot \mathbf{W} \cdot \mathbf{u}|^2 + \sum_a g\left([\mathbf{W} \cdot \mathbf{u}]_a\right) \right\rangle + \ln |\det \mathbf{W}| + K \,, \tag{10.36}$$

where K is independent of $\mathbf{G}$. Under the conventional EM procedure, we would maximize this expression with respect to $\mathbf{G}$, keeping $\mathbf{W}$ fixed. However, the normal procedure fails in this case, because the minimum of the right side of equation 10.36 occurs at $\mathbf{G} = \mathbf{W}^{-1}$, and $\mathbf{W}$ is being held fixed, so $\mathbf{G}$ cannot change. This is an anomaly of coordinate ascent in this particular limit.

Fortunately, it is easy to fix this problem, because we know that $\mathbf{W} = \mathbf{G}^{-1}$ provides an exact inversion of the generative model. Therefore, instead of holding $\mathbf{W}$ fixed during the M phase of an EM procedure, we keep $\mathbf{W} = \mathbf{G}^{-1}$ at all times as we change $\mathbf{G}$. This sets $\mathcal{F}$ equal to the average log likelihood, and the process of optimizing with respect to $\mathbf{G}$ is equivalent to likelihood maximization. Because $\mathbf{W} = \mathbf{G}^{-1}$, maximizing with respect to $\mathbf{W}$ is equivalent to maximizing with respect to $\mathbf{G}$, and it turns out that this is easier to do. Therefore, we set $\mathbf{W} = \mathbf{G}^{-1}$ in equation 10.36, which causes the first term to vanish, and write the remaining terms as the log likelihood expressed as a function of $\mathbf{W}$ instead of $\mathbf{G}$,

$$L(\mathbf{W}) = \left\langle \sum_a g\left([\mathbf{W} \cdot \mathbf{u}]_a\right) \right\rangle + \ln |\det \mathbf{W}| + K \,. \tag{10.37}$$

Direct stochastic gradient ascent on this log likelihood can be performed using the update rule

$$\mathbf{W}_{ab} \to \mathbf{W}_{ab} + \epsilon \left(\left[\mathbf{W}^{-1}\right]_{ba} + g'(v_a) u_b \right) \,, \tag{10.38}$$

where ϵ is a small learning rate parameter, and we have used the fact that $\partial \ln |\det \mathbf{W}| / \partial \mathbf{W}_{ab} = [\mathbf{W}^{-1}]_{ba}$.

The update rule of equation 10.38 can be simplified by using a clever trick. Because $\mathbf{W}^T \mathbf{W}$ is a positive definite matrix (see the Mathematical

Appendix), the weight change can be multiplied by $\mathbf{W}^T\mathbf{W}$ without affecting the fixed points of the update rule. This means that the alternative learning rule

$$\mathbf{W}_{ab} \to \mathbf{W}_{ab} + \epsilon \left(\mathbf{W}_{ab} + g'(v_a)\,[\mathbf{v}\cdot\mathbf{W}]_b\right) \tag{10.39}$$

has the same potential final weight matrices as equation 10.38. This is called a natural gradient rule, and it avoids the matrix inversion of $\mathbf{W}$ as well as providing faster convergence. Equation 10.39 can be interpreted as the sum of an anti-decay term that forces $\mathbf{W}$ away from 0, and a generalized type of anti-Hebbian term. The choice of prior $p[v] \propto 1/\cosh(v)$ makes $g'(v) = -\tanh(v)$ and produces the rule

$$\mathbf{W}_{ab} \to \mathbf{W}_{ab} + \epsilon \left(\mathbf{W}_{ab} - \tanh(v_a)\,[\mathbf{v}\cdot\mathbf{W}]_b\right) . \tag{10.40}$$

This algorithm is called independent components analysis. Just as the sparse coding network is a nonlinear generalization of factor analysis, independent components analysis is a nonlinear generalization of principal components analysis that attempts to account for non-Gaussian features of the input distribution. The generative model is based on the assumption that $\mathbf{u} = \mathbf{G}\cdot\mathbf{v}$. Some other technical conditions must be satisfied for independent components analysis to extract reasonable causes. Specifically, the prior distributions over causes $p[v] \propto \exp(g(v))$ must be non-Gaussian and, at least to the extent of being correctly super- or sub-Gaussian, must faithfully reflect the actual distribution over causes. The particular form $p[v] \propto 1/\cosh(v)$ is super-Gaussian, and thus generates a sparse prior. There are variants of independent components analysis in which the prior distributions are adaptive.

The independent components algorithm was suggested by Bell and Sejnowski (1995) from the different perspective of maximizing the mutual information between $\mathbf{u}$ and $\mathbf{v}$ when $v_a(\mathbf{u}) = f([\mathbf{W}\cdot\mathbf{u}]_a)$, with a particular, monotonically increasing nonlinear function f. Maximizing the mutual information in this context requires maximizing the entropy of the distribution over $\mathbf{v}$. This, in turn, requires the components of $\mathbf{v}$ to be as independent as possible because redundancy between them reduces the entropy. In the case that $f(v) = g'(v)$, the expression for the entropy is the same as that for the log likelihood in equation 10.37, up to constant factors. Thus, maximizing the entropy and performing maximum likelihood density estimation are identical.

One advantage independent components analysis has over other sparse coding algorithms is that, because the recognition model is an exact inverse of the generative model, receptive as well as projective fields can be constructed. Just as the projective field for v_a can be represented by the matrix elements G_{ab} for all b values, the receptive field is given by W_{ab} for all b.

To illustrate independent components analysis, figure 10.8 shows an (admittedly bizarre) example of its application to the sounds created by tapping a tooth while adjusting the shape of the mouth to reproduce a

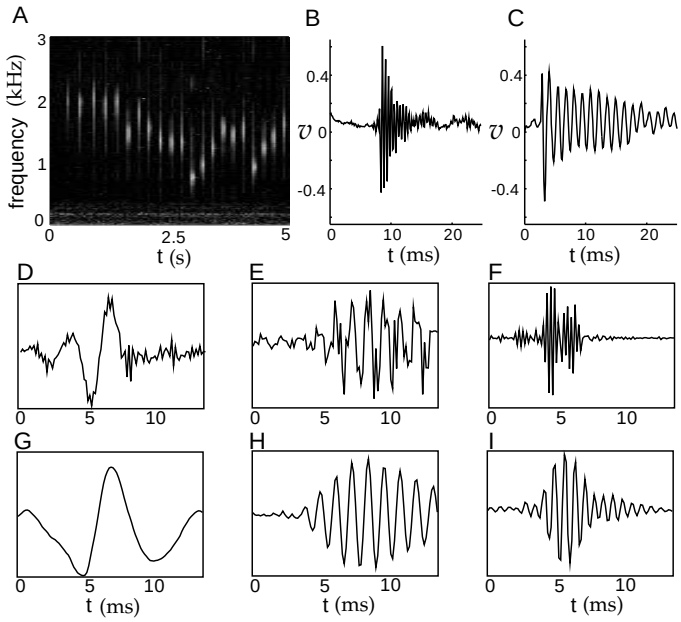

Figure 10.8 Independent components of tooth-tapping sounds. (A) Spectrogram of the input. (B-C) Waveforms for high- and low-frequency notes. The mouth acts as a damped resonant cavity in the generation of these tones. (D-F) Three independent components calculated on the basis of $1/80$ s samples taken from the input at random times. The graphs show the receptive fields (from **W**) for three output units. D is reported to be sensitive to the sound of an air conditioner. E and F extract tooth taps of different frequencies. (G-I) The associated projective fields (from **G**), showing the input activity associated with the causes in D-F. (Adapted from Bell & Sejnowski, 1996.)

tune by Beethoven. The input, sampled at 8 kHz, has the spectrogram shown in figure 10.8A. In this example, we have some idea about likely causes. For example, the plots in figures 10.8B and 10.8C show high- and low-frequency tooth taps, although other causes arise from the imperfect recording conditions (e.g., the background sound of an air conditioner). A close variant of the independent components analysis method described above was used to extract $N_v = 100$ independent components. Figure 10.8D, 10.8E, and 10.8F show the receptive fields of three of these components. The last two extract particular frequencies in the input. Figure 10.8G, 10.8H, and 10.8I show projective fields. Note that the projective fields are much smoother than the receptive fields.

Bell and Sejnowski (1997) also used visual input data similar to those used in the example of figure 10.7, along with the prior $p[v] \propto 1/\cosh(v)$, and found that independent components analysis extracts Gabor-like receptive fields similar to the projective fields shown in figure 10.7A.

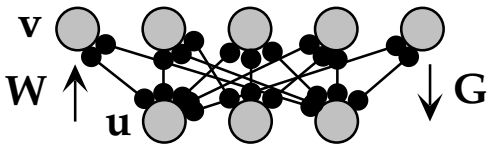

Figure 10.9 Network for the Helmholtz machine. In the bottom-up network, representational units **v** are driven by inputs **u** through feedforward weights **W**. In the top-down network, the inputs are driven by the **v** units through feedback weights **G**.

The Helmholtz Machine

The Helmholtz machine was designed to accommodate hierarchical architectures that construct complex multilayer representations. The model involves two interacting networks, one with parameters $\mathcal{G}$ that is driven in the top-down direction to implement the generative model, and the other, with parameters $\mathcal{W}$, driven bottom-up to implement the recognition model. The parameters are determined by a modified EM algorithm that results in roughly symmetric updates for the two networks.

We consider a simple, two-layer, nonlinear Helmholtz machine (figure 10.9) with binary units, so that u_b and v_a for all b and a take the values 0 or 1. For this model,

$$P[\mathbf{v}; \mathcal{G}] = \prod_a \left(f(g_a)\right)^{v_a} \left(1 - f(g_a)\right)^{1-v_a} \tag{10.41}$$

$$P[\mathbf{u}|\mathbf{v}; \mathcal{G}] = \prod_b \left(f\left(h_b + [\mathbf{G}\cdot\mathbf{v}]_b\right)\right)^{u_b} \left(1 - f\left(h_b + [\mathbf{G}\cdot\mathbf{v}]_b\right)\right)^{1-u_b}, \tag{10.42}$$

where g_a is a generative bias weight for output a that controls how frequently $v_a = 1$, h_b is the generative bias weight for u_b, and $f(g) = 1/(1+\exp(-g))$ is the standard sigmoid function. The generative model is thus parameterized by $\mathcal{G} = (\mathbf{g}, \mathbf{h}, \mathbf{G})$. According to these distributions, the components of **v** are mutually independent, and the components of **u** are independent given a fixed value of **v**.

The generative model is noninvertible in this case, so an approximate recognition distribution must be constructed. This uses a form similar to equation 10.42, except with bottom-up weights **W** and biases **w**,

$$Q[\mathbf{v}; \mathbf{u}, \mathcal{W}] = \prod_a \left(f\left(w_a + [\mathbf{W}\cdot\mathbf{u}]_a\right)\right)^{v_a} \left(1 - f\left(w_a + [\mathbf{W}\cdot\mathbf{u}]_a\right)\right)^{1-v_a} \tag{10.43}$$

The parameter list for the recognition model is $\mathcal{W} = (\mathbf{w}, \mathbf{W})$. This distribution is only an approximate inverse of the generative model because it implies that the components of **v** are independent when, in fact, given a particular input **u**, they are conditionally dependent due to the way they interact in equation 10.42 to generate **u** (this is the same assumption as in the mean-field approximate distribution for the Boltzmann machine, except that the parameters of the distribution here are shared between all input cases).

The EM algorithm for this noninvertible model would consist of alternately maximizing the function $\mathcal{F}$ given by

$$\mathcal{F}(\mathcal{W}, \mathcal{G}) = \left\langle \sum_{\mathbf{v}} Q[\mathbf{v}; \mathbf{u}, \mathcal{W}] \ln \frac{P[\mathbf{v}, \mathbf{u}; \mathcal{G}]}{Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]} \right\rangle \tag{10.44}$$

with respect to the parameters $\mathcal{W}$ and $\mathcal{G}$. For the M phase of the Helmholtz machine, this is exactly what is done. However, during the E phase, maximizing with respect to $\mathcal{W}$ is problematic because the function $Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]$ appears in two places in the expression for $\mathcal{F}$. This also makes the learning rule during the E phase take a different form from that during the M phase. Instead, the Helmholtz machine uses a simpler and more symmetric approximation to EM.

The approximation to EM used by the Helmholtz machine is constructed by re-expressing $\mathcal{F}$ from equation 10.10, explicitly writing out the average over input data and the expression for the Kullback-Leibler divergence,

$$\begin{aligned}\mathcal{F}(\mathcal{W}, \mathcal{G}) &= L(\mathcal{G}) - \sum_{\mathbf{u}} P[\mathbf{u}] D_{\mathrm{KL}}(Q[\mathbf{v}; \mathbf{u}, \mathcal{W}], P[\mathbf{v}|\mathbf{u}; \mathcal{G}]) \\ &= L(\mathcal{G}) - \sum_{\mathbf{u}} P[\mathbf{u}] \sum_{\mathbf{v}} Q[\mathbf{v}; \mathbf{u}, \mathcal{W}] \ln \left(\frac{Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]}{P[\mathbf{v}|\mathbf{u}; \mathcal{G}]} \right) .\end{aligned} \tag{10.45}$$

This is the function that is maximized with respect to $\mathcal{G}$ during the M phase for the Helmholtz machine. However, the E phase is not based on maximizing equation 10.45 with respect to $\mathcal{W}$. Instead, an approximate $\mathcal{F}$ function that we call $\tilde{\mathcal{F}}$ is used. This is constructed by using $P[\mathbf{u}; \mathcal{G}]$ as an approximation for $P[\mathbf{u}]$ and $D_{\mathrm{KL}}(P[\mathbf{v}|\mathbf{u}; \mathcal{G}], Q[\mathbf{v}; \mathbf{u}, \mathcal{W}])$ as an approximation for $D_{\mathrm{KL}}(Q[\mathbf{v}; \mathbf{u}, \mathcal{W}], P[\mathbf{v}|\mathbf{u}; \mathcal{G}])$ in equation 10.45. These are likely to be good approximations if the generative and approximate recognition models are accurate. Thus, we write

$$\begin{aligned}\tilde{\mathcal{F}}(\mathcal{W}, \mathcal{G}) &= L(\mathcal{G}) - \sum_{\mathbf{u}} P[\mathbf{u}; \mathcal{G}] D_{\mathrm{KL}}(P[\mathbf{v}|\mathbf{u}; \mathcal{G}], Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]) \\ &= L(\mathcal{G}) - \sum_{\mathbf{u}} P[\mathbf{u}; \mathcal{G}] \sum_{\mathbf{v}} P[\mathbf{v}|\mathbf{u}; \mathcal{G}] \ln \left(\frac{P[\mathbf{v}|\mathbf{u}; \mathcal{G}]}{Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]} \right) .\end{aligned} \tag{10.46}$$

and maximize this, rather than $\mathcal{F}$, with respect to $\mathcal{W}$ during the E phase. This amounts to averaging the "flipped" Kullback-Leibler divergence over samples of $\mathbf{u}$ created by the generative model, rather than real data samples. The advantage of making these approximations is that the E and M phases become highly symmetric, as can be seen by examining the second equalities in equations 10.45 and 10.46.

Learning in the Helmholtz machine proceeds by using stochastic sampling to replace the weighted sums in equations 10.45 and 10.46. In the M phase, an input $\mathbf{u}$ from $P[\mathbf{u}]$ is presented, and a sample $\mathbf{v}$ is drawn from the current recognition distribution $Q[\mathbf{v}; \mathbf{u}, \mathcal{W}]$. Then the generative weights $\mathcal{G}$ are changed according to the discrepancy between $\mathbf{u}$ and the generative or top-down prediction $\mathbf{f}(\mathbf{h} + \mathbf{G} \cdot \mathbf{v})$ of $\mathbf{u}$ (see the appendix). Thus, the generative model is trained to make $\mathbf{u}$ more likely to be generated by the cause

$\mathbf{v}$ associated with it by the recognition model. In the E phase, samples of both $\mathbf{v}$ and $\mathbf{u}$ are drawn from the generative model distributions $P[\mathbf{v}; \mathcal{G}]$ and $P[\mathbf{u}|\mathbf{v}; \mathcal{G}]$, and the recognition parameters $\mathcal{W}$ are changed according to the discrepancy between the sampled cause $\mathbf{v}$ and the recognition or bottom-up prediction $\mathbf{f}(\mathbf{w} + \mathbf{W} \cdot \mathbf{u})$ of $\mathbf{v}$ (see the appendix). The rationale for this is that the $\mathbf{v}$ that was used by the generative model to create $\mathbf{u}$ is a good choice for its cause in the recognition model.

wake-sleep algorithm

The two phases of learning are sometimes called wake and sleep because learning in the first phase is driven by real inputs $\mathbf{u}$ from the environment, while learning in the second phase is driven by values $\mathbf{v}$ and $\mathbf{u}$ "fantasized" by the generative model. This terminology is based on slightly different principles from the wake and sleep phases of the Boltzmann machine discussed in chapter 8. The sleep phase is only an approximation of the actual E phase, and general conditions under which learning converges appropriately are not known.

10.4 Discussion

Because of the widespread significance of coding, transmitting, storing, and decoding visual images such as photographs and movies, substantial effort has been devoted to understanding the structure of this class of inputs. As a result, visual images provide an ideal testing ground for representational learning algorithms, allowing us to go beyond evaluating the representations they produce solely in terms of the log likelihood and qualitative similarities with cortical receptive fields.

Most modern image (and auditory) processing techniques are based on multi-resolution decompositions. In such decompositions, images are represented by the activity of a population of units with systematically varying spatial frequency and orientation preferences, centered at various locations on the image. The outputs of the representational units are generated by filters (typically linear) that act as receptive fields and are partially localized in both space and spatial frequency. The filters usually have similar underlying forms, but they are cast at different spatial scales and centered at different locations for the different units. Systematic versions of such representations, in forms such as wavelets, are important signal processing tools, and there is an extensive body of theory about their representational and coding qualities. Representation of sensory information in separated frequency bands at different spatial locations has significant psychophysical consequences as well.

The projective fields of the units in the sparse coding network shown in figure 10.7 suggest that they construct something like a multi-resolution decomposition of inputs, with multiple spatial scales, locations, and orientations. Thus, multi-resolution analysis gives us a way to put into sharper focus the issues arising from models such as sparse coding and independent components analysis. After a brief review of multi-resolution decom-